

SMART CITY BIKE SHARING ANALYTICS

Project Overview

This dashboard analyzes bike-sharing station data to monitor bike availability, station utilization, and operational performance across cities.

Key Metrics

- Total Stations: 1,028
- Total Bikes: 5,961
- Total Empty Docks: 10,481
- Bike Availability Rate: 30.7%
- Open Stations: 913 (88.81%)

Key Insights

- Lyon has the highest number of available bikes among all cities.
- Most stations are operational, with 913 stations currently open.
- Bike availability is 30.7%, indicating active usage of the bike-sharing network.
- 73.6% of stations fall under the Low Utilization category.
- Empty docks (10.48K) exceed available bikes (5.96K), showing high bike circulation and demand.
- The geographical distribution shows a strong concentration of bike-sharing stations across European cities.
- Station 10002 - INSA has the highest bike availability among all stations.

Recommendations

- Increase the number of bikes at high-demand stations.
- Rebalance bikes regularly between low-demand and high-demand stations.
- Improve maintenance to reduce station closures.
- Monitor stations with low bike availability to improve service quality.
- Expand bike-sharing services in cities with higher usage.
- Analyze peak usage periods to support better planning and resource allocation.

Conclusion

The bike-sharing network demonstrates strong operational performance with a high percentage of active stations and significant user demand. By improving bike redistribution, maintenance activities, and demand monitoring, the overall efficiency and customer experience of the bike-sharing system can be further enhanced.

